

Technology Stack

Date	27 June 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

The **Smart Lender** project uses a modern AI/ML technology stack to automate loan eligibility prediction and simplify the lending process. Python is used for machine learning model development, while Streamlit provides a simple and interactive user interface. Loan applicant data is securely stored in a database, and Scikit-learn is used to train and evaluate prediction models. GitHub is used for version control and collaboration, making the application scalable, reliable, and easy to maintain.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, Streamlit	Creates a simple, responsive, and user-friendly interface for entering loan details and displaying predictions.
2	Backend / Server-Side	Python, Flask	Handles user requests, processes loan data, and connects the frontend with the ML model.
3	AI / Machine Learning	Scikit-learn, Pandas, NumPy	Used for data preprocessing, model training, and loan eligibility prediction.
4	Database / Data Storage	MY SQL	Stores customer details, loan applications, and prediction history securely.
5	Cloud / Deployment	Streamlit Cloud / Render	Hosts the application online for easy access and future scalability.
6	Version Control	Matplotlib, Joblib	Matplotlib visualizes data, and Joblib saves and loads trained ML models efficiently.